Programming for Problem Solving Using Python
Unit-1

Session 4: Python Syntax, Keywords, linting with flake8/pylint

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Session-4: RECAP

Python Features:-

- Simple and Easy to Learn
- Interpreted Language
- Dynamically Typed
- Extensive Standard Library
- Supports Multiple Paradigms
- Strong Community and Ecosystem
- Robust Frameworks and Libraries
- Integrated Development Environments (IDEs)

Session 4:Why Use VS code For Python?

- Lightweight & powerful IDE
- Built-in terminal and Git support
- IntelliSense for code completion
- Interactive debugging with breakpoints
- Support for virtual environments

Linting with flake8 and pylint

Linting = Catch Errors Early

flake8: Finds style issues and unused imports
pylint: More detailed, checks for code smells

Installation:

```
pip install flake8 pylint
```

Run Linter:

```
flake8 my_script.py
pylint my_script.py
```

Formatting with black

black = The Uncompromising Code Formatter

Automatically formats code to follow PEP 8

Fast, opinionated, and no arguments about style

Installation:

```
pip install black
```

Usage:

```
black myscript.py
```

Getting started with Python in Python Online Text Editor

Your First Program after Python Installation

As a developer you write Python (.py) files in a text editor and then put those files into the python interpreter to be executed. The way to run a python file is like this on the command line:

C:\Users\Your Name> python helloworld.py

Where "helloworld.py" is the name of your python file.

First Python Program in Python Online Text Editor

Let's write our first Python file, called helloworld.py, which can be done in any text editor.

helloworld.py

```
Print("Hello, World!")
```

Save your file. Open your command line, navigate to the directory where you saved your file, and run:
C:\Users\Your Name>python helloworld.py

OUTPUT:- Hello, World!

Python Interactive Shell

- Start the Python interpreter by simply typing python (or python3 on some systems) in the terminal or com- mand prompt. We can then run Python commands interactively.
- To exit, we can type exit() or press Ctrl + D.

```
PS C:\Users\kamal> python
Python 3.11.4 (tags/v3.11.4:d2340ef, Jun  7 2023, 05:45:37) [MSC v.1934 64 bi
t (AMD64)] on win32
Type "help", "copyright", "credits" or "license" for more information.
>>> print("Namaste Geeks!")
Namaste Geeks!
>>> exit()
PS C:\Users\kamal>
```

Python Interactive Shell

Running a Python Script

- Create a script with the .py extension, e.g., Namaste.py.

- Run it from the terminal or command prompt using: python script.py (or python Namaste.py).

```
PS D:\Office\Python> python Namaste.py
Namaste Geeks
PS D:\Office\Python>
```

figures/KRWU-

Python Command Line

in Shell

To test a short amount of code in python sometimes it is quickest and easiest not to write the code in a file. This is made possible because Python can be run as a command line itself. Type the following on the Windows, Mac or Linux command line:

```
C:\Users \Your Name>python
```

Or, if the ”python” command did not work, you can try ”py”:

```
C:\Users \Your Name>py
```

Python Command Line (contd.)

in Shell

From there you can write any python, including our hello world example from Slide-104:

```
C:\Users\Your Name>python
Python 3.6.4 (v3.6.4:d48eceb, Dec 19 2017, 06:04:45) [MSC v.1900 32 bit (Intel)] on win32
Type "help", "copyright", "credits" or "license" for more information.
>>> print("Hello, World!")
```

Which will write ”Hello, World!” in the command line:

```
C:\Users\Your Name>python
Python 3.6.4 (v3.6.4:d48eceb, Dec 19 2017, 06:04:45) [MSC v.1900 32 bit (Intel)] on win32
Type "help", "copyright", "credits" or "license" for more information.
>>> print("Hello, World!")
Hello, World!
```

Python Command Line (contd..)

Whenever you are done in the python command line, you can simply type the following to quit the python command line interface:

```
exit()
```

Python Syntax

Python syntax can be executed by writing directly in the Command Line:

```
>>> print("Hello, World!")
Hello, World!
```

Or by creating a python file on the server, using the .py file extension, and running it in the Command Line:

```
C:\Users\Your Name>python myfile.py
```

Python Lines and Indentation

- Python programming provides no braces to indicate blocks of code for class and function definitions or flow control.
- Blocks of code are denoted by line indentation, which is rigidly enforced.
- Indentation refers to the spaces at the beginning of a code line.
- Where in other programming languages the indenta- tion in code is for readability only, the indentation in Python is very important.

Python Lines and Indentation (contd.)

- Python uses indentation to indicate a block of code. The
- number of spaces in the indentation is variable, but all statements within the block must be indented the same amount.
- Example:-

Example

```
if 5 > 2:
  print("Five is greater than two!")
```

Python Lines and Indentation (contd..)

- Python will give you an error if you skip the indentation:

Example

Syntax Error:

```
if 5 > 2:
print("Five is greater than two!")
```

- The number of spaces is up to you as a programmer, the most common use is four, but it has to be at least one.

Example

```
if 5 > 2:
 print("Five is greater than two!")
if 5 > 2:
        print("Five is greater than two!")
```

Python Comments

- Comments in Python are the lines in the code that are ignored by the interpreter during the execution of the program.
- Comments enhance the readability of the code and help the programmers to understand the code very carefully.
- It also helps in collaborating with other developers as adding comments makes it easier to explain the code.

Python Comments (contd.)

Just like any other language, comments in Python serve following purpose:

- Enhance code readability
- Explaining code to other
- Understanding code if studied after some time
- Documenting the steps and needs for a function
- Sharing code with fellow developers Collaborating
- with multiple people.

Python Comments (contd..)

Python supports these types of comments:

- **Single Line Comments:** In a Python script, the symbol # marks the beginning of comment line. It is effective till the end of line in the editor. If # is the first character of line, then entire line will be assumed as a comment and interpreter will ignore it.

```
# This is a comment
print ("Hello World")
```

```
# This is a comment
# print ("How are you?")
print ("Hello World")
```

figu

Python Comments (contd...)

- **Multi Line Comments:** In Python, there is no provision to write multi-line comment, or a block comment. Each line should have the # symbol at the start to be marked as comment in Python and that's how you can create multi-line comments in Python. Many Python IDEs have shortcuts to mark a block of statements as comment.

```
#
# This is a multi-line comment
#  which can span through multi-line.
#
print ("Hello World")
```

figures/IK

Python Comments (contd....)

- **Multi Line Comments:** A triple quoted multi-line string is also treated as comment if it is not a doc-string of a function or class.

```
'''
First line in the comment
Second line in the comment
Third line in the comment
'''
print ("Hello World")
```

Python Comments (contd....)

- **Docstring Comments:** Python docstring is the string literals with triple quotes that are appeared right after the function. It is used to associate documentation that has been written with Python modules, func- tions, classes, and methods.

```
def add(a, b):
    """Function to add the value of a and b"""
    return a+b

print(add.__doc__)
```

Python Comments (contd....)

- **Docstring Comments:** It is added right below the functions, modules, or classes to describe what they do. In Python, the docstring is then made available via the doc__attribute.

```
def multiply(a, b):
    """Multiplies the value of a and b"""
    return a*b

# Print the docstring of multiply function
print(multiply.__doc__)
```

Output:

```
Multiplies the value of a and b
```

Python Empty Lines

An empty line is a blank line that is composed of spaces, tab spaces.

```
1.  print("Hello World")
2.  print()
3.  print("I am a Python Programmer")
```

```
1.  print("Hello World")
2.  print('')
3.  print("I am a Python Programmer")
```

```
1.  print("Hello World")
2.  print("")
3.  print("I am a Python Programmer")
```

figu

Python Blank Lines

- In Python, a blank line is essentially an empty line placed between two segments of text. It helps in presenting the output in a clean and comprehensible manner. The newline character \n is commonly used to create such blank lines.
- We can print multiple blank lines in python by using the \n character the number of times we need a blank line.

Python Blank Lines (contd.)

For example, If you need 5 blank lines, you can use:-

```
Python 2.x:
print "\n\n\n\n\n"

Python 3.x:
print("\n\n\n\n\n")
```

You can use features like repetition operator in python to make this easier. For example,

```
Python 2.x:
print "\n" * 5

Python 3.x:
print("\n" * 5)
```

Python Quotations

- There are different types of quotes in Python. Each quotation is used in different scenarios as per the requirement.
- Python accepts single (’), double (”) and triple (”’ or ”””) quotes to denote string literals, as long as the same type of quote starts and ends the string.
- The triple quotes are used to span the string across multiple lines. Example:-

```
word = 'word'
sentence = "This is a sentence."
paragraph = """This is a paragraph. It is
made up of multiple lines and sentences."""
```

Python Quotations (contd.)

- **Single quotes:-** Single quotes (' ') are used to cre- ate strings and characters in Python programming. Since the print() function accepts string values we can directly pass values to it wrapping them within the single quotes.
- **Double quotes:-** We can also use the Double quotes (" ") to create strings and characters. The function- ality of single (') and double (") quotes is the same; you can use either one depending on your require- ments.

Triple quotes:- The Triple quotes are used for com-

- menting and representing the docString in the python.

Python Multi-line statements

- New line indicate end of statement in python but if you want a statement to continue in a new line, then the line continuation character '\' can be used to denote the line continues in the new line.
- If the '\' operator is written at the end of a line, Python interpreter automatically goes to the next line as a continuation to the statement. This is known as explicit line continuation.

Python Multi-line statements (contd.)

Example 1:-

```
item_one = 1
item_two = 3
item_three = 4

total = item_one + \
        item_two + \
        item_three

print (total)
```

Example 2:-

```
total = 12 + \
        22 + \
        33
print(total)
```

figures/KR

Python Multi-line statements (contd..)

- In Python, you can freely break lines inside parenthe- ses (( ), { }, [ ]). You can use this rule to split a long string into multiple lines using parentheses instead of backslashes.
- This is known as implicit line continuation and python interpreter will understand them all.

Python Multi-line statements (contd...)

```
total = (12 + \
         22 + \
         33)

print(total)

total = {13 + \
         65 + \
         19}

print(total)

total = [45 + \
         6 + \
         77]
print(total)
```

Output

On executing the program above, the result is produced as –

```
67
{97}
[128]
```

Python Keywords

" Python keywords are reserved words that are used to define the syntax and structure of the Python language. They are the fundamental elements that form the 'grammar' of Python code."

Python Keywords

- **if**: Conditional statement for executing code based on a condition.
- **else**: Executes when the if condition is false.
- **elif**: Short for "else if", used for multiple branching in conditions.
- **for**: Used for looping over a sequence (e.g., list, tuple, string).
- **while**: Loops until a condition becomes false.

Python Keywords (contd.)

- **def**: Defines a function.
- **return**: Exits a function and returns a value.
- **class**: Defines a new class.
- **import**: Imports a module into the current script or module.
- **try**, **except**: Used for exception handling.

Python Keywords (contd.)

```
import keyword
# printing all keywords at once using "kwlist()"
print("The list of keywords is : ")
print(keyword.kwlist)
```

Python Program i- Use of Keywords

Python program to illustrate the use of keywords.
Define a function to check if a number is even or odd.

```
def check_even_odd(num):
    if num % 2 == 0:
        print(f"{num} is even")
    else:
        print(f"{num} is odd")

# Test the function
check_even_odd(5)
check_even_odd(10)
```

Python Program ii- Use of Keywords

```
# Python program to illustrate use of keywords.
# Define a function to calculate the factorial of a number.
```

```
def factorial(n):
    if n == 0:
        return 1
    else:
        return n * factorial(n-1)
# Test the function
number = 5
fact = factorial(number)
print("Factorial of", number, "is", fact)
```

Python Program iii- Use of Keywords

```
# Python program to illustrate use of keywords.
# Define a function to find the maximum of two numbers.
```

```
def find_maximum(a, b):
    if a > b:
        return a
    else:
        return b
# Test the function
x = 5
y = 10
print("The maximum of", x, "and", y, "is:",
find_maximum(x, y))
```

Python Program iii- Use of Keywords

This output demonstrates the use of a function named find_maximum to find the maximum of two numbers. In this case, the function is called with arguments (5, 10), and it correctly determines that 10 is greater than 5, returning 10 as the maximum value.

OUTPUT:

The maximum of 5 and 10 is: 10

Python Program iv: Using `try` and `except`

Python program to illustrate the use of try-except

```
def divide(x, y):
    try:
        result = x / y
        print("Result of division:", result)
    except ZeroDivisionError:
        print("Error: Division by zero!")

# Test the function
divide(10, 2)
divide(5, 0)
```

Python Program iv: Using `try` and `except`

In the first call to the divide function with arguments (10, 2), the division operation 10 / 2 succeeds, and the result 5.0 is printed. In the second call with arguments (5, 0), a ZeroDivisionError occurs due to division by zero, and the exception is caught by the except block, printing the error message "Error: Division by zero!".

```
Result of division: 5.0
Error: Division by zero!
```

Python Program v: Using for and while

Python program to illustrate the use of for and while loops

```
# Using for loop to iterate over a sequence
fruits = ["apple", "banana", "cherry"]
print("Using for loop:")
for fruit in fruits:
    print(fruit)

# Using while loop to iterate until a condition
becomes false
count = 0
print("\nUsing while loop:")
while count < 5:
    print("Count:", count)
    count += 1
```

figures/KRM

Session-3: Python Program v: Using `for` and `while`

Using for loop:
apple
banana
cherry

Using while loop:
Count: 0
Count: 1
Count: 2
Count: 3
Count: 4

Python Program vi: Using class

Python program to illustrate the use of class keyword

```
class Dog:
  # The init method or constructor
    def __init__(self, name, breed):
        self.name = name
        self.breed = breed
    def bark(self):
        print(f"{self.name} barks!")
# Creating an instance of the Dog class
my_dog = Dog("Buddy", "Golden Retriever")
# Accessing attributes and calling methods of
the instance
print(" Name:", my_dog.name)
print(" Breed:", my_dog.breed)
my_dog.bark()
```

figures/KRM

Python Program vi: Using `class`

OUTPUT:

```
Name: Buddy
Breed: Golden Retriever
Buddy barks!
```

Test Your Knowledge- Fill in the blanks

- Which keyword is used to define a function in Python?

  _____

- Which symbol is used for string concatenation in Python?

  _____

- In Python, what does the range() function return? _____
- Which keyword is used to exit a loop prematurely in Python?

  _____

- What does the pass keyword do in Python? _____
- How to declare a variable in Python? _____

Test Your Knowledge- Fill in the blanks (Solutions)

- Which keyword is used to define a function in Python? def
- Which symbol is used for string concatenation in Python? +
- In Python, what does the range() function return?
  A generator object
- Which keyword is used to exit a loop prematurely in Python?
  break
- What does the pass keyword do in Python?
  It is a placeholder indicating "do nothing."
- How to declare a variable in Python? my_variable = 10

Test Your Knowledge- Quiz I

- What is the correct syntax for defining a function in Python?

  1. def my_function():
  2. function my_function():
  3. define my_function():
  4. my_function def():

- What is the correct way to start a for loop in Python?

  1. for i in range(10):
  2. for i = 0; i < 10; i++:
  3. foreach i in range(10):
  4. loop i in range(10):

- How do you define a list in Python?

  1. [1, 2, 3]
  2. {1, 2, 3}
  3. (1, 2, 3)
  4. "1, 2, 3"

Test Your Knowledge- Quiz I (Solutions)

- What is the correct syntax for defining a function in Python?
  1. def my_function():
  2. function my_function():
  3. define my_function():
  4. my_function def():
- What is the correct way to start a for loop in Python?
  1. for i in range(10):
  2. for i = 0; i < 10; i++:
  3. foreach i in range(10):
  4. loop i in range(10):
- How do you define a list in Python?
  1. [1, 2, 3]
  2. {1, 2, 3}
  3. (1, 2, 3)
  4. "1, 2, 3"

Test Your Knowledge- Quiz II

- Which of the following is a correct way to write a single-line if statement in Python?

  1. if x == 5 then print("x is 5")
  2. if x == 5 print("x is 5")
  3. if x == 5; print("x is 5")
  4. if x == 5: print("x is 5")

- What is the correct syntax for a while loop in Python?

  1. while (condition)
  2. while condition:
  3. while (condition):
  4. while condition

Test Your Knowledge- Quiz II (Solutions)

- Which of the following is a correct way to write a single-line if statement in Python?

  1. if x == 5 then print("x is 5")
  2. if x == 5 print("x is 5")
  3. if x == 5; print("x is 5")
  4. if x == 5: print("x is 5")

- What is the correct syntax for a while loop in Python?

  1. while (condition)
  2. while condition:
  3. while (condition):
  4. while condition

Test Your Knowledge- Interview Questions

1. Differentiate between == and 'is' in Python.
2. How do you comment multiple lines of code in Python?
3. What is the significance of the __init__ method in Python classes?
4. How do you define a function with an arbitrary number of arguments in Python?

Test Your Knowledge- Interview Questions (Solutions)

1. Ans:- == is the equality operator in Python, used to compare the values of two objects. It checks if the values of the two operands are equal.
   is is the identity operator in Python, used to compare the memory addresses (identities) of two objects. It checks if two operands refer to the same object in memory.

2. Ans:- In Python, you can use triple quotes (''' or """) to create multi-line comments.

Test Your Knowledge- Interview Questions (Solutions)

3. Ans:- The __init__ method is a special method in Python classes that is automatically called when a new instance of the class is created.
   Ans:- It initializes (sets up) the object's initial state by assigning values to its attributes. It's commonly used to initialize instance variables and perform any necessary setup actions when creating objects.

4. Ans:- In Python, you can define a function with an arbitrary number of arguments using the *args and **kwargs syntax.

Thank You!

Any Questions?